

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 05



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Recap of Previous Lecture



✓
Topic

Transitive relation

✓
Topic

Reflexive closure

✓
Topic

Symmetric closure

✓
Topic

Transitive closure

✓
Topic

Complement of a relation and Inverse of a relation

Topics to be Covered



✓ **Topic**

Composite of two relations

✓ **Topic**

Equivalence relation

✓ **Topic**

Equivalence class

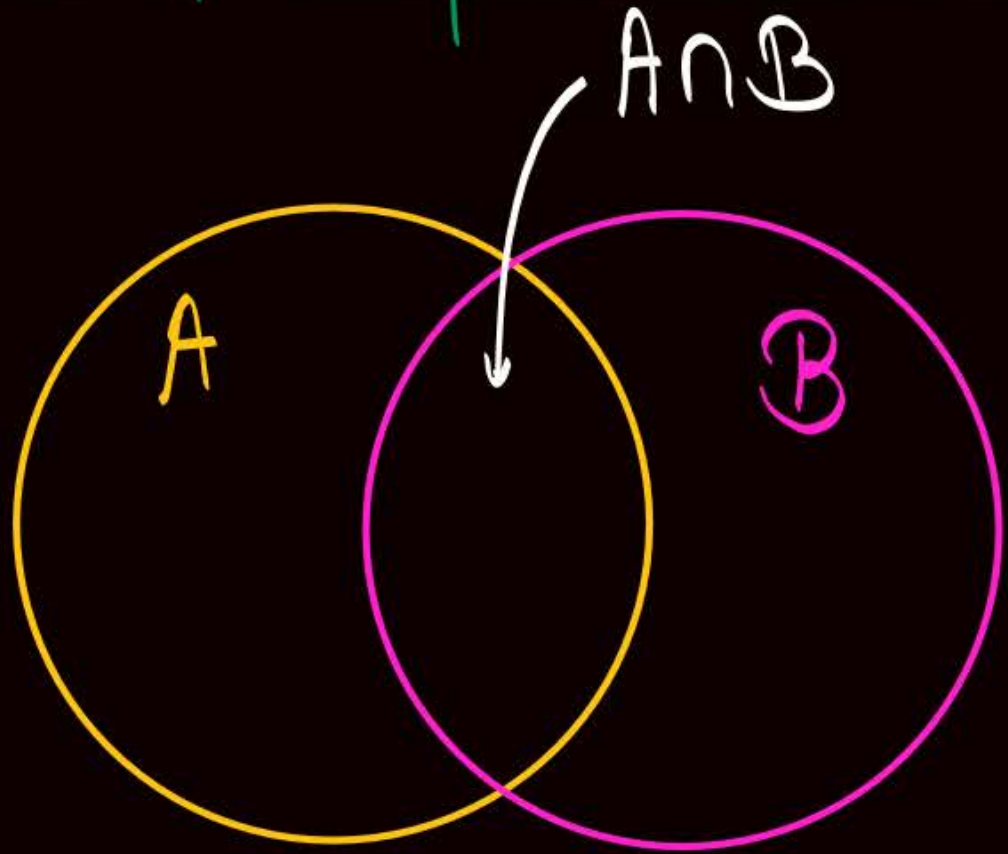
✓ **Topic**

Partition of a set

✓ **Topic**

Number of partitions of a set

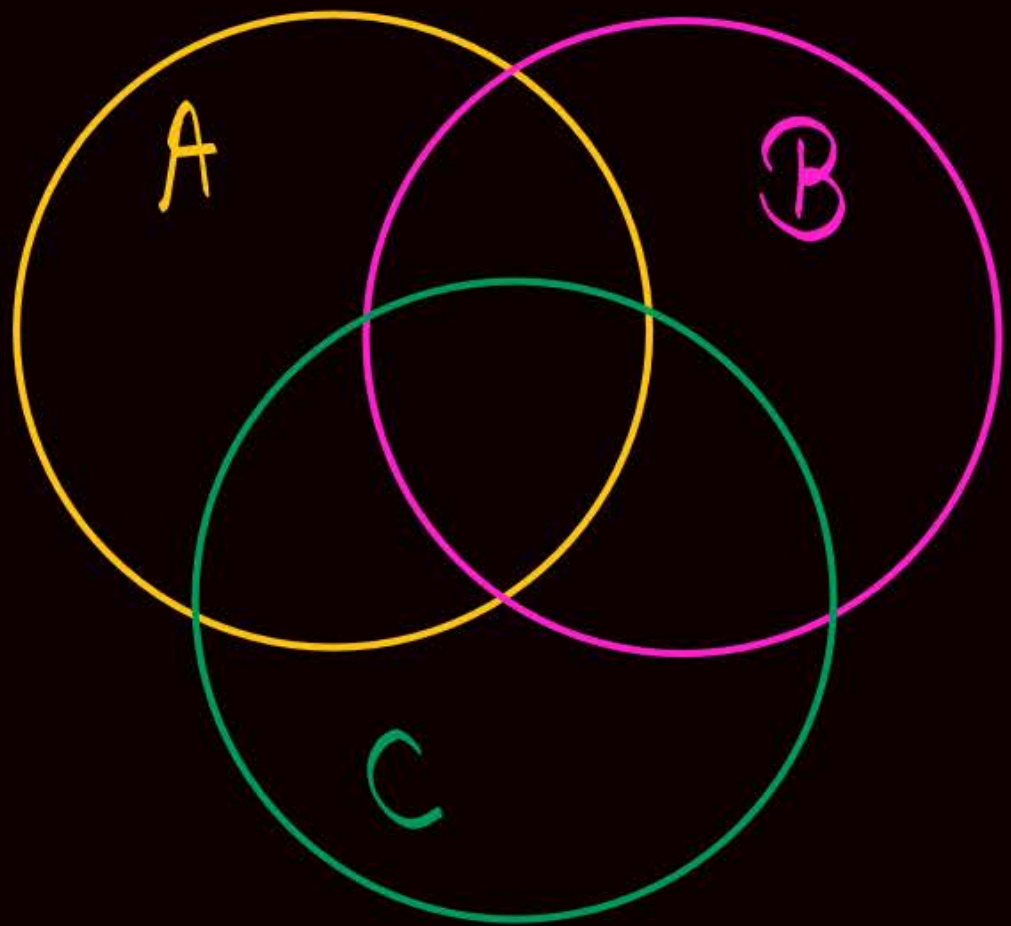
Principle of inclusion & exclusion :-



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Principle of inclusion & Exclusion :-



$$\begin{aligned}n(A \cup B \cup C) &= n(A) + n(B) + n(C) \\&\quad - n(A \cap B) - n(A \cap C) - n(B \cap C) \\&\quad + n(A \cap B \cap C)\end{aligned}$$

14.W. Q. Let X and Y are two sets such that

$$|X^c| = 15$$

$$|Y^c| = 8$$

and $|X \cap Y^c| = 5$

$$\begin{aligned} |X^c \cup Y^c| &= \\ &= |X^c| + |Y^c| - |X^c \cap Y^c| \\ &= 15 + 8 - 3 = \underline{\underline{20}} \end{aligned}$$

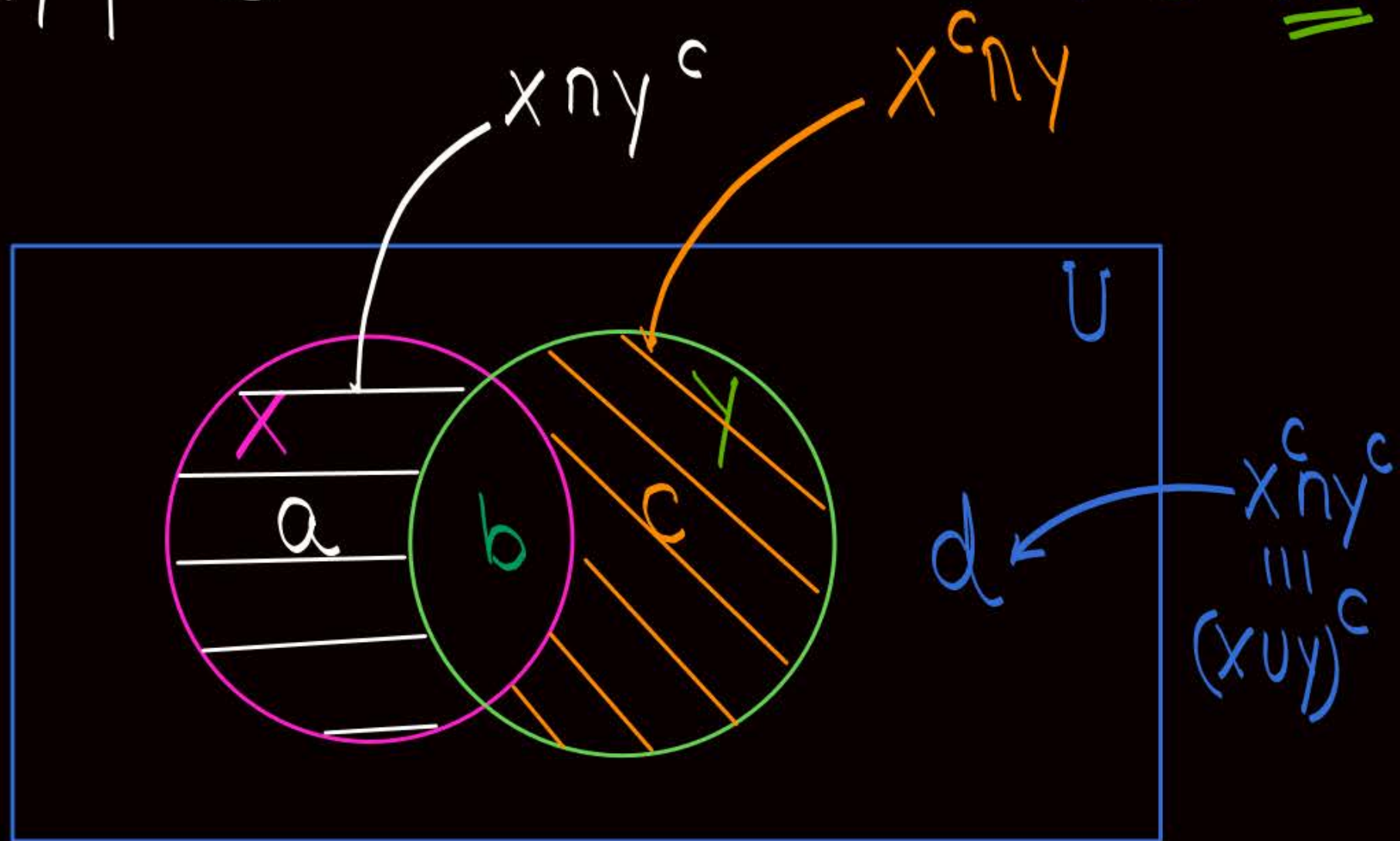
then $|X^c \cap Y| = ? = \boxed{C = 12}$

$$|X^c| = c + d = 15 \text{ — ①}$$

$$|Y^c| = a + d = 8 \text{ — ②}$$

$$|X \cap Y^c| = a = 5 \text{ — ③}$$

by ② & ③ $\boxed{d = 3}$ — ④ By ① & ④ $\boxed{C = 12}$



H.W

Q. The expression

$$(A \cup B \cup C') \cap ((A' \cap B' \cap C) \cup C')$$

is equivalent to

$$\{(A \cup B)' \cap C\} \cup C'$$

(a) $(A \cup B) - C'$

(b) $(A \cup B) \cap C$

(c) $(A \cup B) \cap C'$

☒ (d) None of the above

$$\{(A \cup B)' \cup C'\} \cap (\underbrace{C \cup C'}_U)$$

$$((A \cup B)' \cup C')$$

$$((A \cup B) \cup C') \cap ((A \cup B)' \cup C')$$
$$((A \cup B) \cap (A \cup B)') \cup C'$$

$$\emptyset \cup C' \equiv C'$$

H.W.
MSQ
Q.:

Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule

$(a, b) \in R$ if and only if blood group of a is same as blood group of b

then Relation R is/an

- ☒ (a) Reflexive
- ☒ (b) Symmetric
- ☒ (c) Transitive
- ☒ (d) Anti-symmetric

H.W MSQ Q.:

Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule $(a, b) \in R$ if and only if 'a' is brother of 'b'

{ R will be irreflexive

then Relation R is

- ☒ Reflexive
- ☒ Symmetric
- ☒ Transitive
- ☒ Anti-symmetric
- ☒ None of the above

① No person will be brother of himself $\therefore$ Not Reflexive

② if Ram is brother of Geeta ^{Male} _{Female}, but Geeta will not be brother of Ram
i.e. if $(\text{Ram}, \text{Geeta}) \in R$ then $(\text{Geeta}, \text{Ram}) \notin R$

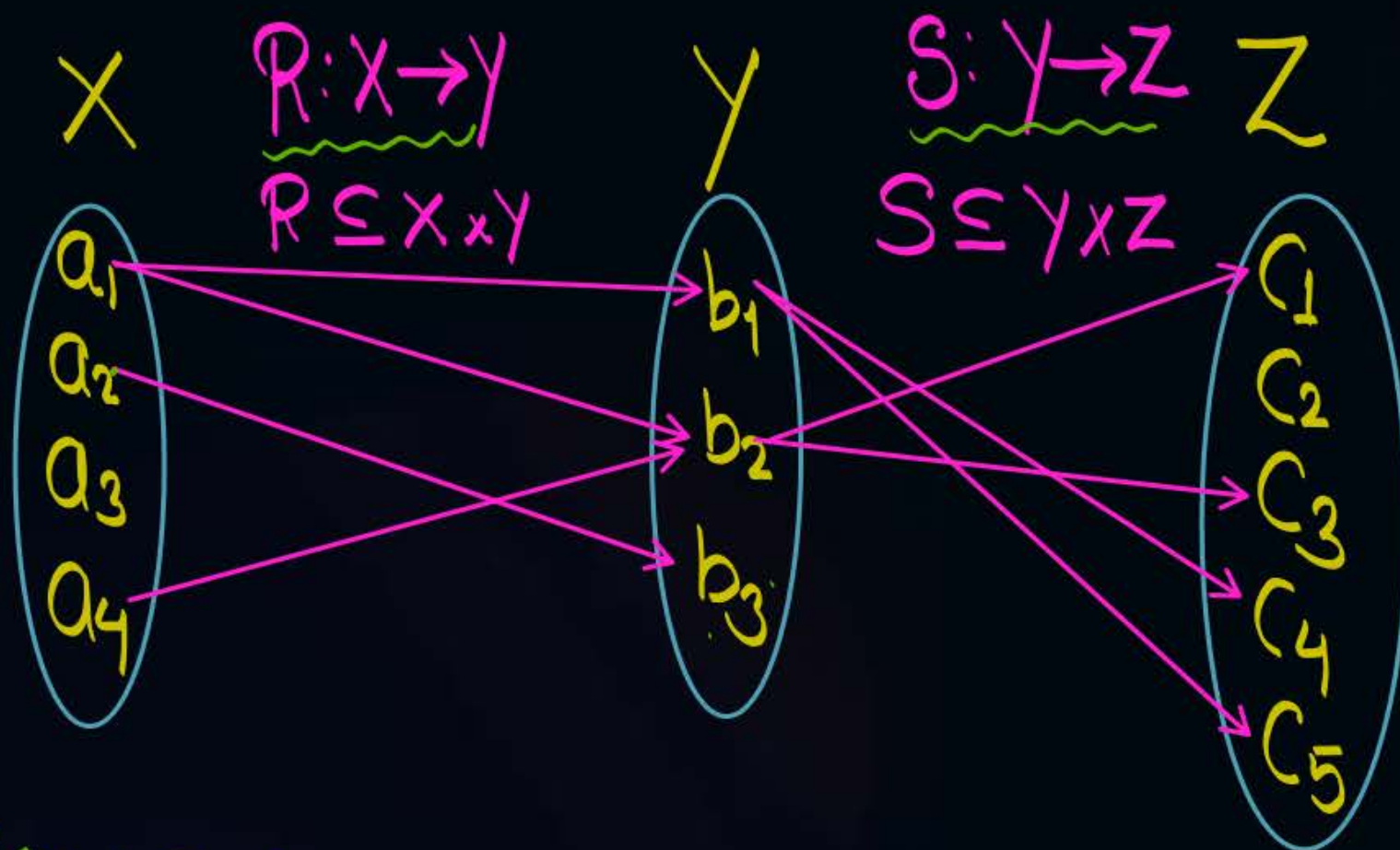
④ let $(\text{Ram}, \text{Shyam}) \in R$, then $(\text{Shyam}, \text{Ram}) \in R$
^{Male} _{Male} $\therefore$ Not anti-symmetric

③ From ④ $(\text{Ram}, \text{Shyam}) \in R$ & $(\text{Shyam}, \text{Ram}) \in R$

$\therefore (\text{Ram}, \text{Ram})$ must be present in R
but $\therefore$ Not transitive



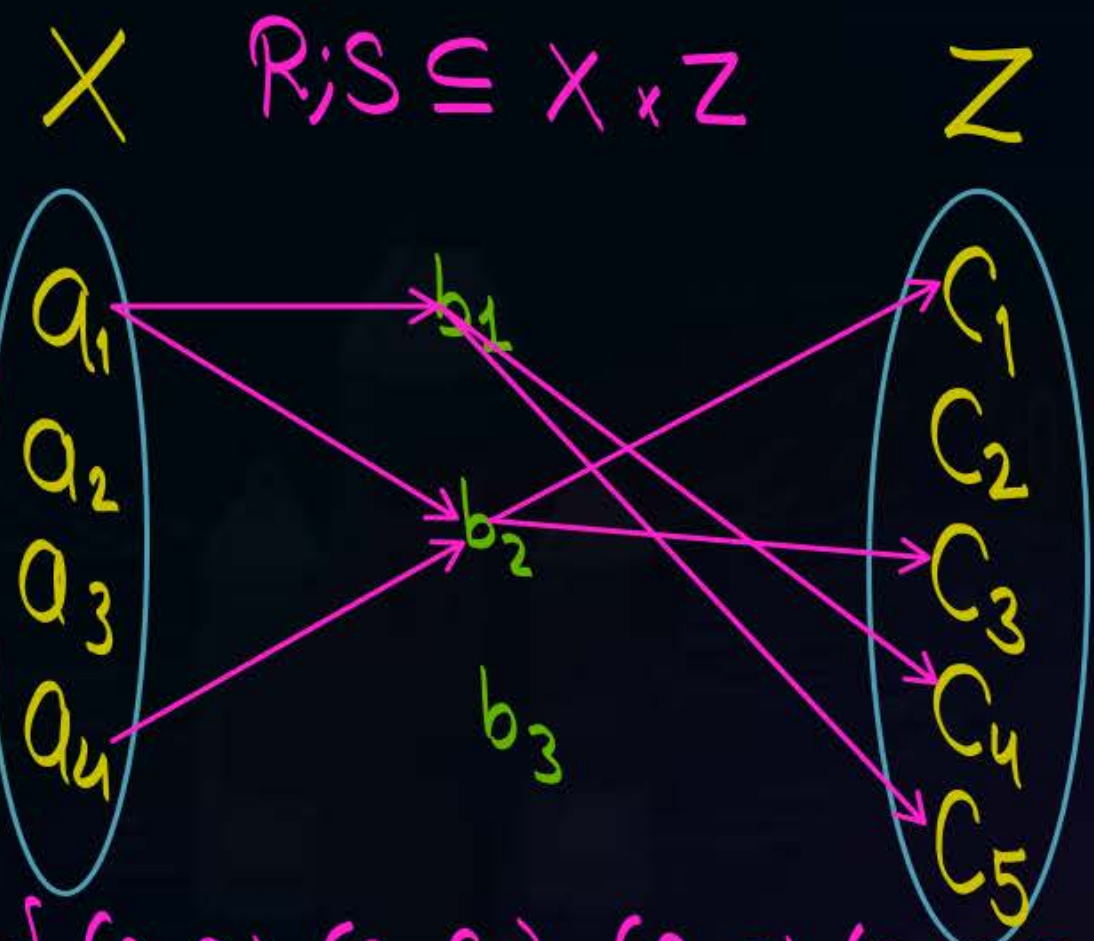
Topic : Composite of two relations



- * $R = \{(a_1, b_1), (a_1, b_2), (a_2, b_2), (a_3, b_2), (a_4, b_2)\}$
- * $S = \{(b_1, c_4), (b_1, c_5), (b_2, c_1), (b_2, c_2), (b_2, c_3)\}$

Composite of relation
 R & S i.e. $R \circ S$
is a relation from X to Z

$R \circ S : X \rightarrow Z$



$R \circ S = \{(a_1, c_4), (a_1, c_5), (a_1, c_1), (a_1, c_3), (a_4, c_1), (a_4, c_3)\}$



Topic : Composite of two Relations

Let X, Y and Z be three sets, and

If R is a relation from X to Y , and S is a relation from Y to Z ,
then their composition $R;S$ is the relation defined as,
or simply "RS"

$$R;S = \{ (a,c) \mid (a,c) \in X \times Z \text{ and there exists } b \in Y \text{ such that } (a,b) \in R \text{ and } (b,c) \in S \}$$

ie. $R;S$ will be a relation from X to Z

i.e., $R;S$ is a relation from X to Z defined by the rule that $(a,c) \in R;S$ if and only if there is an element $b \in Y$ such that $(a,b) \in R$ and $(b,c) \in S$



Topic : Equivalence relation

• A relation R on set A is called an Equivalence relation if and only if relation R is

✓ ① Reflexive
and ✓ ② Symmetric
and ✓ ③ Transitive

Equivalence Relation = Reflexive ✓ + Symmetric ✓ + Transitive ✓

eg. let $A = \{1, 2, 3\}$

$$R_1 = \{(1,1), (2,2), (3,3)\} = \Delta_A$$

Reflexive
Symmetric
Transitive

✓
✓
✓
} $\circ \circ$ Equivalence Relation

Δ_A is the smallest Equivalence rel^n on set A

$$R_2 = "A \times A" = \left. \begin{array}{l} \text{Reflexive} \\ \text{Symmetric} \\ \text{Transitive} \end{array} \right\} \circ \circ \text{Equivalence Rel}^n$$

$'A \times A'$ is the largest Equivalence rel^n on set A

Which of the following are Equivalence relⁿ on set A
eg. let $A = \{1, 2, 3\}$

$$R_3 = \{ \underbrace{(1,1), (2,2), (3,3)}, \underbrace{(1,2), (2,1)} \}$$

Reflexive ✓
Symmetric ✓
Transitive ✓

} Equivalence Relⁿ

$$R_4 = \{ \underbrace{(1,1), (2,2), (3,3)}, \underbrace{(1,3), (3,1)} \}$$

Reflexive ✓
Symmetric ✓
Transitive ✓

} Equivalence Relⁿ

$$R_5 = \{ \underbrace{(1,1), (2,2), (3,3)}, \underbrace{(2,3), (3,2)} \}$$

Reflexive ✓
Symmetric ✓
Transitive ✓

} Equivalence Relⁿ

eg. Let $A = \{1, 2, 3\}$

Symmetric ✓

Reflexive ✓

$$R_6 = \{ (1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1) \}$$

∴ By transitivity $(2,3)$ must be present in R_6

But $(2,3) \notin R_6$ ∴ Not transitive

Hence not an
Equivalence Relⁿ

→ Check whether the statement is true or false.

"Every symmetric as well as transitive relation on set 'A', is also a reflexive relation on set A"

eg let $A = \{1, 2, 3\}$

$R = \{ \}$

Empty relation
on set A

← it is symmetric as well as transitive
but it is not reflexive

eg:

let $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (2,2), (1,2), (2,1)\}$

→ Symmetric + Transitive

But it is not reflexive



Topic : Equivalence Class

- Let R is an Equivalence relation on set A .
- Equivalence class of an element $x \in A$ wrt. Equivalence relation R can be denoted by $[x]$ and it is defined as,

$$[x] = \{ y \mid x R y \}$$

(or)

$$[x] = \{ y \mid (x, y) \in R \}$$

eg: Let $A = \{1, 2, 3, 4, 5\}$
and R is an Equivalence relation on set A .

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

Find Equivalence class of every element of set A
w.r.t. Equivalence relation R .

Soln.

$$[1] = \{1, 2\}$$

$$[2] = \{2, 1\} = \{1, 2\}$$

$$[3] = \{3\}$$

$$[4] = \{4, 5\}$$

$$[5] = \{5, 4\} = \{4, 5\}$$

Same Equivalence class for
element '1' & '2'

$$\text{i.e. } [1] = [2]$$

Three distinct
Equivalence classes
for elements of set A .

Same Equivalence class
for '4' & '5'

$$\text{i.e. } [4] = [5]$$

→ Distinct Equivalence classes of elements
of set A are $\{1, 2\}$, $\{3\}$, $\{4, 5\}$

→ Set of all distinct Equivalence
Class of elements of set A w.r.t.
Equivalence Relⁿ ' R ' defines the
Partition of set ' A ' w.r.t. Equivalence Relⁿ ' R '
i.e. $\{\{1, 2\}, \{3\}, \{4, 5\}\}$ is a partition
of set A w.r.t. Equivalence Relⁿ ' R '

→ Note :- → ① Equivalence class of two distinct elements of the set may be equal eg:- $[1] = [2]$ even when $1 \neq 2$

Note :- ② Set of all distinct Equivalence classes of elements of set A will be a partition of set A with respect to given Equivalence relation



Topic : Partition of a set

→ Partition of a "set A" is the set of non-empty subsets of set A such that each element of set A appears in exactly one non-empty subset of that set (or)

→ Let A is any set and $A_1, A_2, A_3, \dots, A_k$ are non-empty subsets of set A; then

$\{A_1, A_2, A_3, \dots, A_k\}$ is a partition of set A if and only if

(i) $A_i \cap A_j = \emptyset \quad \forall i, j (i \neq j)$
and (ii) $\bigcup_{i=1}^k A_i = A$

eg:- Let $A = \{1, 2, 3, 4, 5\}$
Which of the following is/are partitions of set A.

(a) $\{\{1, 2, 3\}, \{3, 4\}, \{5\}\}$
 $\cap = \{3\} \neq \emptyset \therefore$ Not a Partition

(b) $\{\{1, 2\}, \{3\}, \{4\}\} \Rightarrow$ Union is not equal to 'A' $\therefore$ Not a partition

(c) $\{\{1, 2, 3\}, \{4\}, \{5\}, \{\}$ } Empty subset is not allowed in a set w.o.t. partition

(d) $\{\{1, 2\}, \{3\}, \{4\}, \{5\}\}$

(e) $\{\{1, 2, 3, 4, 5\}\}$

(f) $\{\{1, 2\}, \{3\}, \{4, 5\}\}$

→ Find the number of partitions of "set A" when

(1) $|A| = 0$

(2) $|A| = 1$

(3) $|A| = 2$

(4) $|A| = 3$

(5) $|A| = 4$

① Let $A = \{ \}$, i.e. $|A| = 0$

Partition of set $A = \{ \}$

Partition of set A
will be a set

→ and

elements of this set
will be non-empty subsets of set A

→ If set A itself is empty, then
no non-empty subset of set A .

∴ No element in the set w.r.t.
Partition of set A .

Partition itself may be
an empty set,
but an empty set can
never be an element of
the set w.r.t. Partition

i.e.,

If $|A| = 0$ then number of partitions of set $A = 1$

② Let $A = \{a\}$, i.e. $|A| = 1$

$$|A| = 1 \Rightarrow$$

1
only one subset
of size = 1

Partition of set $A =$

Partition of set A
will be a set

$\{ \{a\} \}$

↑
and elements will be
non-empty subsets of set A

w. a set of Cardinality '1', there will be only one
non-empty subset, and it will be set A itself.

i.e., If $|A| = 1$ then number of partitions of set $A = 1$

$$\begin{array}{c} \text{No. of Equivalence Rel}^n \\ \text{On set A} \end{array} = \begin{array}{c} \text{No. of Partitions} \\ \text{of set A} \end{array}$$



2 mins Summary



- ✓ **Topic** Composite of two relations
- ✓ **Topic** Equivalence relation
- ✓ **Topic** Equivalence class
- ✓ **Topic** Partition of a set
- ✓ **Topic** Number of partitions of a set

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THANK - YOU